



Microlearning and generative AI for pre-service teacher education: a qualitative case study

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Abstract

The rapid emergence of generative artificial intelligence (GenAI) tools has underscored the urgent need for pre-service teachers to develop technological pedagogical content knowledge (TPACK) and self-regulated learning (SRL) strategies – both critical for integrating AI into classrooms. However, existing teacher education programmes lack structured approaches to equip pre-service teachers with AI literacy and pedagogical adaptation skills. Traditional training models remain too generalised and fail to provide incremental, hands-on experiences for AI integration. This qualitative case study addresses these gaps by investigating the use of microlearning modules – bite-sized, multimodal instructional units – to enhance pre-service teachers’ TPACK and SRL in English language teaching (ELT). Over 13 weeks, 19 participants engaged with GenAI-focused microlearning modules that progressively developed their ability to adapt AI tools such as ChatGPT, Twee, Mizou, Perplexity and MagicSchool for differentiated instruction, formative assessment and culturally responsive teaching. Thematic analysis of participants’ reflective journals and semi-structured interviews revealed three key findings: (1) microlearning facilitated a structured, low-cognitive-load approach to developing GenAI competencies, (2) participants gained confidence and autonomy in using AI for lesson planning, and (3) SRL strategies such as goal setting and iterative refinement were essential for AI integration. Participants mitigated challenges such as GenAI tool limitations and initial AI anxiety by refining prompt engineering techniques and cross-validating AI outputs. These findings highlight microlearning’s potential to bridge AI literacy and pedagogical applications of AI, offering a scalable model for teacher education.

Keywords Microlearning · Generative artificial intelligence · Teacher education · Pre-service teachers

1 Introduction

The rapid development of artificial intelligence (AI) has created a vital need for pre-service teachers to develop technological pedagogical content knowledge (TPACK; Kohnke & Zou, 2025; Stockwell & Wang, 2023; Zou et al., 2022). Effective teacher education is essential to equip future educators with digital and pedagogical skills for navigating the evolving landscape of generative AI (GenAI) to improve student outcomes and overall school performance (Darling-Hammond et al., 2017). However, teacher preparation today remains too generalised and disconnected from the practical needs of future classrooms (Celik et al., 2022; Cullen & Green, 2011). Traditional programmes often follow outdated ‘one-size-fits-all’ models that fail to address pre-service teachers’ specific challenges, and provide structured, hands-on experiences with emerging technologies like GenAI (Ding et al., 2024; Moorhouse & Kohnke, 2024). Therefore, more flexible, personalised, and targeted learning approaches that promote self-regulated learning (SRL) are needed, as GenAI requires distinct competencies and pedagogical strategies that differ significantly from traditional digital tools (Chiu, 2024).

Microlearning offers a promising solution, delivering content in bite-sized, multimodal segments (1–6 min; Kohnke, 2023; Reinhardt & Elwood, 2019). This approach is particularly suited for teaching complex technologies like GenAI because it allows pre-service teachers to progressively develop their understanding at their own pace without feeling overwhelmed (Kohnke et al., 2024a). By structuring content into small, digestible units, microlearning can enhance knowledge retention and promote continuous adaptation by breaking complex topics into manageable lessons (Zhang & West, 2020). It also facilitates personalized SRL, with learners controlling their pace and learning paths (Shamir-Inbal & Blau, 2022).

Moreover, microlearning’s alignment with the TPACK framework enables pre-service teachers to engage in hands-on, practical applications of AI technologies that integrate technological, pedagogical, and content knowledge (TPCK) (Mishra & Koehler, 2006). For example, microlearning modules on GenAI tools can enhance language teachers’ technological knowledge (TK) by equipping them with the skills to effectively use AI for language teaching tasks, such as generating prompts or creating assessments. Moreover, designing such tasks requires the application of teachers’ content knowledge (CK), which simultaneously cultivates their technological content knowledge (TCK) as they integrate AI capabilities with subject-specific expertise. These modules also reinforce technological pedagogical knowledge (TPK) by guiding teachers in applying AI tools in ways that enhance teaching strategies, such as differentiated instruction or formative assessment. In this way, TPCK is enhanced as pre-service teachers adapt AI tools to teacher-specific language skills or content areas, ensuring that the technology supports students’ effective achievement of subject-specific learning goals. By delivering content in short, manageable modules, microlearning promotes both technological fluency and pedagogical integrity, helping teachers develop the skills they need to integrate GenAI into their teaching practice (Jahnke et al., 2020; Pölzl-Stefanec & Geißler, 2022).

However, some scholars argue that microlearning, while effective for short-term learning gains, might not always facilitate deep learning or higher-order thinking capabilities (Rahman et al., 2021; Rof et al., 2024). Short learning modules may limit opportunities for sustained reflection and in-depth pedagogical exploration (Buchem & Hamelmann, 2010). Furthermore, as microlearning emphasise self-directed learning, pre-service teachers unfamiliar with AI tools may struggle with metacognitive regulation (Kohnke et al., 2024a). Despite these concerns, empirical research on the effectiveness of microlearning in preparing pre-service teachers to manage GenAI integration is limited (Kohnke & Moorhouse, 2024).

To address this gap, this study explores how microlearning modules focused on GenAI tools can enhance not only TPACK development but also SLR strategies, offering new insights into how emerging technologies can transform teacher education. Specifically, we investigate the following research questions (RQ):

RQ 1: How do pre-service teacher perceive the role of microlearning modules focused on GenAI tools in developing their TPACK knowledge and SLR strategies in lesson planning and instruction?

RQ 2: How do pre-service teachers engage with and apply microlearning-based GenAI training in their teaching practice?

RQ 3: What challenges and benefits do pre-service teachers experience when using microlearning modules to learn and implement GenAI tools, particularly in relation to SRL strategies as goal setting, reflection, and metacognitive monitoring?

2 Literature review

2.1 Microlearning in teacher education

Microlearning uses technology to create and distribute digestible ‘chunks’ of content, focusing on a single concept and incorporating multimodal features like animation and infographics (Hug, 2005; Kohnke, 2023). It allows teachers to work at their own pace, reducing their cognitive loads (Corbeil et al., 2021), enriching the learning process (Jomah et al., 2016) and improving knowledge retention (Kohnke et al., 2024b; Zhang & Wester, 2020). Microlearning aligns with the development of SRL strategies, enabling pre-service teachers to actively manage their learning processes and navigate the challenges of integrating GenAI (Shamir-Inbal & Blau, 2022). As an instructional strategy, microlearning is particularly effective for complex topics like AI literacy, enabling learners to engage with content incrementally and reducing cognitive overload. The present study defines microlearning activities as lasting 2–8 min and using technology to provide learners with personalised learning content and experience (Kohnke & Moorhouse, 2024).

Research has shown that microlearning is effective in various educational contexts, including teacher education. Reinhardt and Elwood (2019) and Kohnke et al. (2024a, b) reported that microlearning enhanced the digital literacy of pre-service teachers by simplifying the acquisition of complex technology skills. Kohnke and

Moorhouse (2024) found that using microlearning to self-direct continuing professional development among in-service teachers allowed them to gradually build technological competence. AI tools provided real-time feedback and adapted instructional content to individual learning needs, fostering deeper engagement with TPACK.

Although microlearning is effective for skill acquisition, some scholars have raised concerns about its limitations in fostering deeper pedagogical engagement (Rahman et al., 2021; Rof et al., 2024). A major concern is that fragmented, short-duration learning experiences may encourage surface-level engagement rather than the sustained inquiry needed for deep learning (Buchem & Hamelmann, 2010). Because microlearning segments are brief, learners may not have adequate opportunities to critically reflect on pedagogical applications or synthesise concepts across modules (Alias & Razak, 2025; Rof et al., 2024). Recent research suggests that AI-driven reflection tools (automated teaching simulations, multimodal teaching behaviour analysis) can mitigate this issue by prompting learners to engage in structured and deeper analysis of pedagogical strategies (Tan et al., 2025; Phillips et al., 2022; Zheng et al., 2024).

This assertion highlights a gap in understanding how microlearning should be structured to support TPACK development. Some scholars suggest that microlearning should not function as a stand-alone approach but instead be integrated into broader learning frameworks that include reflective and collaborative elements (Kohnke & Zou, 2025). Embedding structured reflection exercises—such as requiring learners to critically evaluate AI-generated materials or discuss ethical considerations—could enhance the pedagogical impact of microlearning by fostering critical AI literacy alongside technical competence (Pack & Maloney, 2023). Similarly, integrating peer discussions and guided mentoring into microlearning-based training programmes may mitigate the risk posed by fragmented learning experiences (Rahman et al., 2021).

2.2 GenAI in education

GenAI refers to AI systems (e.g. ChatGPT, Claude) capable of producing new content according to user input, such as text, images, audio and code. In English language teaching (ELT), GenAI tools like MagicSchool and Alayna can automatically create personalised learning materials, provide real-time feedback and streamline grading processes (Chiu, 2024; Ulla et al., 2023). However, the practical application of these tools in teacher education and their integration into pre-service teachers' pedagogical practices require further research (Moorhouse, 2024).

Teacher professional development (TPD) aims to enhance educators' knowledge and skills (Desimone, 2009; Kennedy, 2016). However, TPD design features alone do not guarantee positive outcomes, as designing learning experiences requires a holistic approach (Kennedy, 2016; Opfer & Pedder, 2011). Initial language teacher education (ILTE) has struggled to keep pace with technological advancements, resulting in difficulties updating accredited programmes and a lack of technical competence (Clark & Newberry, 2019; Ngao et al., 2022; Park & Son, 2022; Starkey,

2020). Such conditions can leave pre-service teachers feeling underprepared to use technology (Farjon et al., 2019; Gudmundsdottir & Hatlevik, 2018). Therefore, it is essential to explore microlearning as a means of developing pre-service teachers' TPACK and SRL strategies.

Although GenAI technologies are increasingly used in classrooms (Kohnke et al., 2023), their role in teacher education continues to evolve (Moorhouse & Kohnke, 2024). Integrating AI into effective classroom teaching to achieve Intelligent-TPACK presents significant challenges in mastering the comprehensive integration of TPACK components (Celik, 2023). It requires understanding the capabilities and limitations of GenAI tools and the pedagogical strategies needed to leverage them effectively for successful attainment of subject-specific learning objectives (Moorhouse, 2024). Pre-service teachers must learn to critically evaluate AI-generated content and adapt it to their teaching contexts and learning objectives (Pack & Maloney, 2023).

Given the rapid evolution of GenAI, some researchers argue that teacher education programmes must go beyond tool-based training to develop critical AI literacy (Celik, 2023; Luckin et al., 2022). This entails equipping teachers with the ability to critically evaluate AI-generated content, addresses biases, and explore ethical considerations in AI-supported learning contexts (Kohnke & Zou, 2025; Pack & Maloney, 2023). Without these advanced competencies, pre-service teachers risk achieving only superficial familiarity with AI tools rather than the pedagogical expertise necessary for meaningful classroom integration (Moorhouse, 2024).

AI has significantly influenced the implementation of instructional pedagogy (Luckin et al., 2022). To harness AI's potential in education, teachers must understand the pedagogical benefits of AI-based tools (Xu, 2020) and possess adequate pedagogical knowledge to use them effectively (Cavalcanti et al., 2021). Teachers with a deeper understanding of AI are better equipped to choose suitable tools that enable personalised learning and prompt feedback (Edwards et al., 2018, 2024; Popenici & Kerr, 2017).

2.3 AI-empowered personalized learning

Recent developments in AI have transformed the landscape of personalised learning by enabling the creation of highly data-driven and adaptive educational environments. Rooted in cognitive and constructivist theories, personalised learning tailors instructional approaches to accommodate individual learners' prior knowledge, cognitive capacities, learning needs, and preferences (Xie et al., 2019). AI-powered personalised learning systems utilise machine learning algorithms, natural language processing, and intelligent data analytics to track learners' progress, infer preferences, and predict optimal learning trajectories (Alserhan et al., 2023). Such systems incorporate intelligent tutoring systems and adaptive learning platforms, delivering customised content, timely feedback, and scaffolding based on each learner's performance and evolving needs (Essa et al., 2023).

Alserhan et al. (2023) investigated how integrating a third-generation learning management system into a personal learning environment could enhance learners'

self-regulation. This personalised learning approach encouraged learners to collect, create, and organise personalised learning tools, fostering autonomy and self-regulated learning. The results, analysed via structural equation modelling, showed that the LMS-supported PLE improved learners' autonomy, self-reflection, and active learning. In another study, Wang et al. (2023) evaluated the effectiveness of the adaptive learning system Squirrel AI Learning in China by comparing it to traditional large-group and small-group classroom instruction. Conducted with eighth-grade students from two provinces, the research shows that students using the adaptive system achieved significantly higher gains in mathematics than those taught by expert teachers in conventional settings. The findings highlight the potential of adaptive learning systems to improve student outcomes and support further investigation into their implementation in China's education system.

AI-driven personalised learning systems can leverage various parameters to optimise educational experiences, including learning performance, content difficulty, learning preferences, and learner perceptions (Essa et al., 2023; Li et al., 2021). By analysing large-scale learner data, these systems can adjust content sequencing, recommend appropriate learning resources, and provide individualised feedback, contributing to meaningful learning (Edwards et al., 2024). Looking forward, the evolution of AI technologies will continue to push the boundaries of personalisation, offering educators powerful tools to support diverse learner needs in increasingly dynamic and flexible learning environments.

2.4 TPACK in teacher Education

Several studies have applied the TPACK framework in teacher education to explore how pre-service teachers develop these interconnected knowledge domains. For example, Lachner et al. (2021) found that targeted training programmes significantly improved pre-service teachers' TPACK towards using educational technology and technology pedagogy. Similarly, Valtonen et al. (2023) found that PK is central to developing TPACK for pre-service teachers, with CK playing a lesser role. These studies highlight the importance of integrating PK with TK in teacher education programmes.

Other research has examined the specific strategies within TPACK-based teacher training programmes. For instance, Koh et al. (2014) demonstrated that collaborative learning activities (e.g. co-creating technology-integrated lesson plans) successfully bolstered advanced pre-service teachers' TPACK development. Building on this finding, Chai et al. (2013) underscored that scaffolding strategies (reflective practices and explicit instruction on the interplay between technology, pedagogy and content) enabled pre-service teachers to critically evaluate and refine their technology use in teaching. Similarly, Mouza et al. (2014) reported that TPACK-based workshops combining theoretical instruction with hands-on practice helped pre-service teachers develop subject-specific technology-integration skills. Lachner and colleagues (2021) confirmed that pre-service teachers attending TPACK module courses acquired more TPACK than those without the modules.

Additionally, scholars have identified problem-based learning as useful for fostering TPACK in teacher education. Koh and Chai (2016) found that engaging teachers in authentic, technology-driven teaching scenarios enhanced their ability to align technology with pedagogical goals while building confidence in digital tools. These findings suggest that teacher education programmes integrating collaborative, reflective and practice-oriented strategies within the TPACK framework prepare pre-service teachers to meet the demands of 21 st-century classrooms more successfully.

2.5 SRL and teacher education

SRL has remained a critical focus in teacher training programmes, as fostering SRL in students requires teachers to develop self-regulation skills and instructional strategy competencies (Backers et al., 2024; Karlen et al., 2023; Ortube et al., 2024). SRL is the process by which learners actively control their learning through goal setting, self-monitoring and reflection (Zimmerman, 2008). Scholars have often conceptualised it as a cyclical model comprising forethought, performance and self-reflection (Zimmerman, 2010), where learners set goals, evaluate progress and adjust their learning strategies. Research has highlighted that teacher training programmes well equip educators with tools to promote SRL among their students. For instance, Kramarski and Michalsky (2009) found that teacher education programmes emphasising metacognitive training significantly enhanced teachers' abilities to design SRL-supportive learning environments. Similarly, Panadero's (2017) review underscored that explicit instruction in SRL models, such as Zimmerman's cyclical learning phases, improves teachers' abilities to scaffold student learning processes.

Moreover, SRL is critical in technology-enhanced learning environments, where learners must acquire new skills and manage their learning processes in dynamic, often autonomous, digital spaces. Research has shown that digital tools can promote SRL, giving learners more control over their learning environments, providing immediate feedback and enabling flexibility in accessing and applying content (Chiu, 2024). For example, Sharmi-Inbal and Blau (2022) found that technology-enhanced microlearning environments allowing learners to set goals, track progress and receive real-time feedback can significantly enhance SRL strategies. AI-powered learning environments further enhance SRL by providing intelligent tutoring, adaptive feedback, and real-time learning analytics that help pre-service teachers refine their regulation strategies (Kohnke & Moorhouse, 2024). These findings suggest that integrating SRL-focused training into teacher education programmes is essential for fostering effective teaching practices that empower students as autonomous, lifelong learners in traditional and technology-enhanced learning environments. In addition, recent research highlight that AI-driven SRL interventions, such as AI-generated progress tracking and automated goal-setting recommendation, can provide personalised scaffolding for pre-service teachers, ensuring continuous skill development (Kohnke et al., 2025).

These studies highlight how technology-enhanced learning environments (like microlearning) can support SRL. However, a limited understanding remains of how to best structure microlearning environments to support ongoing reflection and

adaptation in teacher education – particularly regarding complex and rapidly evolving technologies like GenAI tools. While AI-enhanced SLR tools offer promising benefits, concerns regarding over-reliance on algorithmic feedback and the potential reduction of human-driven critical reflection must be considered (Kohnke & Zou, 2025; Pack & Maloney, 2023). Scholars emphasise that AI should serve as a scaffold rather than a replacement for independent critical thinking teacher education (Seo et al., 2024; Sperling et al., 2024).

This gap proves relevant because developing SRL strategies is crucial for cultivating pre-service teachers into reflective practitioners who adapt to new technologies and effectively integrate them into their teaching practices (Ortube et al., 2024). As AI continues to (re)shape educational landscapes, it is essential to explore how AI-driven SRL strategies can be design to promote both autonomy and deep pedagogical engagement. This study investigates pre-service teachers' perceptions of the challenges and benefits of using microlearning modules to harness GenAI tools. It aims to contribute to the understanding of how microlearning can support SRL skill development in the context of emerging technologies like GenAI in teacher education programmes.

3 Methodology

3.1 Research context and participants

We conducted the study at an English-medium instruction university in Hong Kong during the 2024 fall semester. The participants included 19 undergraduate English-language education students (15 female, 4 male) aged 20 to 23 years in their third or fourth year of a five-year Bachelor of Education programme, specialising in English-language teaching. This study adopts a case study approach (Yin, 2018) to gain an in-depth understanding of how pre-service teachers develop their TPACK and SRL strategies through the integration of GenAI tools in an ELT context.

Participants were recruited through purposive sampling, as they were already enrolled in the elective course, 'Integrating ICT in ESL/EFL Classroom'. The course provides pre-service teachers opportunities to explore innovative teaching and learning activities and applications to support English-language teaching and learning. Before this course, participants had completed their basic pedagogical training and a short teaching practicum experience. However, they reported minimal exposure to GenAI tools in educational contexts, ensuring that their engagement with these tools was largely new. The course met weekly for a 3-h session, with microlearning modules integrated as a core curriculum component.

The sample size of 19 participants was appropriate for a qualitative case study, allowing for in-depth data collection and thematic analysis while maintaining manageable data saturation (Guest et al., 2006). While this qualitative study does not aim for statistical generalizability, it provides valuable insights into the experience of pre-service teachers in a technology-enhanced learning environment. We informed

Table 1 Weekly microlearning modules and tasks

Week	Focus Area	Module Content	Tasks	TPACK Component
1	GenAI Foundations	• Introduction to ChatGPT interface • Basic prompting for ELT capabilities	• Generate simple vocabulary exercises • Create basic grammar activities • Test language-focused prompts	TCK: Using GenAI tools for basic language tasks TCK: Grammar and vocabulary focus TPCK: Designing language-focused prompts
2–4	Language Teaching & Applications	• ELT lesson planning • Language learning materials • Four skills development activities • Assessment in ELT	• Design communicative activities • Construct reading comprehension materials • Develop listening exercises • Generate writing prompts	TPCK: Integrating GenAI into lesson design TPCK: Creating GenAI-generated content for language learning TPCK: Using GenAI to support language skills TPCK: Developing assessments using GenAI Tools
5–8	Advanced ELT Applications	• Differentiated language instruction • Feedback on student writing • Grammar explanation prompts • Cultural content integration	• Create multilevel language tasks • Generate writing feedback • Develop grammar practice activities • Design cultural awareness materials	TPCK: Adapting GenAI to students' proficiency levels TPCK: Providing automated feedback using GenAI TPCK: Designing grammar-focused tasks with GenAI support TCK: Using GenAI for culturally relevant content
9–12	ELT Integration & Evaluation	• Adapting materials for various levels • ELT teaching strategies • Error-correction approaches	• Modify content for proficiency levels • Construct rubrics for language assessment • Develop error-correction activities	TPCK: Adjusting GenAI tools to meet diverse student needs TPCK: Designing rubrics and assessments using GenAI tools TPCK: Using GenAI for error correction strategies

Table 1 (continued)

Week	Focus Area	Module Content	Tasks	TPACK Component
13	Synthesis & Reflection	• Assessment design	• Plan integrated skills lessons	TPCK: Designing integrated skills lessons using GenAI
		• ELT best practices with GenAI	• Final reflection on ELT applications	TPCK: Reflecting on GenAI's role in ELT
		• Future applications in language teaching	• Assemble an ELT resource portfolio	TPCK: Planning future use of GenAI in ELT
		• Implementation planning	• Plan GenAI integration in language classes	TPCK: Planning for GenAI integration

Each 1–6-min microlearning module focused on ELT applications, followed by practical tasks and reflection activities aligned with TESOL/TEFL principles

all participants of the study's purpose, and they signed consent forms. To protect their anonymity, we used pseudonyms throughout the study.

3.2 Learning design and implementation

We conducted this study over 13 weeks during the 2024 fall semester. Each week, participants engaged in a microlearning module (1–6 min) focusing on different aspects of GenAI tools in ELT. We designed the modules to develop participants' TPACK while progressively promoting SRL strategies. The curriculum was structured into five phases, moving from foundational knowledge to advanced applications in ELT contexts, as Table 1 illustrates. Each module included specific tasks encouraging hands-on experience with GenAI tools, particularly focusing on practical applications for ELT. The weekly progression enabled participants to build their competency systematically, starting with essential GenAI functions and advancing to more complex pedagogical applications in language teaching.

The microlearning modules were structured to align with the TPACK framework, ensuring that pre-service teachers developed a balanced understanding of how to integrate TK, PK, and CK into their teaching practice. By grounding the microlearning tasks in TPACK, we ensured that participants not only learned how to use GenAI tools but also understood how these tools could enhance teaching strategies and language content. This design helped pre-service teachers apply TPACK in meaningful ways, enabling the effective use of GenAI in language instruction.

By the end of the 13-week course, participants had developed a comprehensive understanding of how to use GenAI tools to enhance their TPACK and apply these tools in practical teaching scenarios. The microlearning format allowed participants to engage with the content at their own pace, promoting SRL and reflective practice. This approach not only helped participants develop technological fluency but also ensured they could apply these tools in a way that aligns with their pedagogical objectives and content-specific teaching strategies.

3.3 Data collection

We applied a two-phase approach to comprehensively understand participants' experiences with GenAI tools. First, participants completed weekly reflective journals to capture their immediate learning experiences (Dunlap, 2006). Next, they participated in semi-structured interviews to explore their overall development and perspectives in greater depth (Creswell, 2007).

Phase 1: Weekly Reflective Journals.

Throughout the 13-week course, participants maintained structured reflective journals to document their experiences with GenAI tools, learning strategies and teaching applications. Guided prompts encouraged detailed narratives, focusing on the following:

- Learning experiences with the weekly modules
- Application of GenAI tools in ELT contexts
- Challenges encountered and solutions developed
- SRL strategies employed
- Plans for future implementation

Participants submitted their journals weekly, allowing us to continually track how their use of GenAI tools for language teaching developed. Submitting the journals every two weeks allowed us to monitor their growth (e.g. Moon, 2006) using an iterative analytic approach.

Phase 2: Semi-Structured Interviews.

After course completion, we conducted individual semi-structured interviews (41–57 min) with all participants. The interview guide was developed via a preliminary analysis of the reflective journals and aimed to explore the following:

- Overall experiences with the microlearning modules
- Development of TPACK in GenAI applications
- Growth in SRL strategies
- Plans for implementing GenAI in future teaching practice
- Challenges and solutions in using GenAI for ELT

We conducted all interviews in English and audio-recorded and transcribed them verbatim. The transcriptions were shared with the participants for a preliminary member check (Merriam & Tisdell, 2016).

3.4 Data analysis

The data from the reflective journals and the semi-structured interviews were analysed following Braun and Clarke's (2006) six-step thematic analysis approach. We conducted analyses in two phases aligned with our data collection methods. We maintained a meticulous audit trail documenting all research decisions and processes (i.e. detailed notes about data collection methods, coding schemes, theme development and adjustments made during the study).

Phase 1: Analysis of the Reflective Journals.

1. Familiarisation: We immersed ourselves in the journal entries through repeated reading, noting down initial ideas.
2. Initial coding: We systematically coded key segments related to GenAI use, TPACK development and SRL.
3. Theme searching: We collated initial codes into potential themes reflecting participants' experiences and development.
4. Theme review: We checked the themes against the coded extracts and the entire dataset to ensure coherence.
5. Theme definition: We generated clear definitions and names for each theme.

6. Report production: We selected compelling extract examples to support the analysis.

Phase 2: Analysis of the Interview Data.

1. Familiarisation: We read the transcripts multiple times and noted down initial patterns.
2. Initial coding: We performed systematic coding to both confirm themes from Phase 1 and identify new insights.
3. Theme searching: We organised codes into themes, considering their relationship with the journal findings.
4. Theme review: We refined the themes by comparing Phase 1 results and theoretical frameworks.
5. Theme definition: We defined and named the final themes, integrating insights from both data sources.
6. Report production: We selected representative quotes to illustrate key findings.

To ensure systematic and reliable data analysis, we conducted an interim re-coding process two weeks after the initial coding (Anney, 2014) to verify consistency. During this process, each researcher revisited the dataset independently to reduce the risk that preconceptions from the first coding round would be perpetuated. We then conducted a consistency check, comparing the new codes with the originals to identify discrepancies or refinements. A collaborative discussion was held to resolve any inconsistencies and finalise the coding framework, ensuring coherence across the data sets.

To further ensure trustworthiness, we conducted inter-rater reliability checks. A subset of the data was coded independently by each researcher, and discrepancies were discussed until a consensus was reached before applying the final coding scheme. In addition, we implemented a follow-up member check to maintain transparency and trustworthiness (Merriam & Tisdell, 2016). Participants were given copies of the themes, subthemes, and illustrative quotes. No participants requested changes or offered additional suggestions. Through this iterative process, we analysed emergent themes related to TPACK frameworks and SRL theory to understand participants' development in using GenAI tools for ELT. This verification-driven approach ensured that our thematic analysis was systematic, transparent, and valid.

4 Findings

Analysis of weekly reflective journals and post-course interviews revealed three interconnected themes that address our research questions. Theme 1 explores pre-service teachers' perceptions of the microlearning modules supported their development of TPACK and SRL strategies in lesson planning and instruction (RQ1). Theme 2 examines how they engaged with and applied GenAI tools through microlearning within their teaching practice (RQ2). Theme 3 investigates the challenges

and benefits of microlearning for GenAI integration, particularly regarding self-regulated learning strategies such as goal setting, reflection, and metacognitive monitoring (RQ3).

4.1 Theme 1: Dual development of TPACK and SRL through GenAI-focused microlearning

Our analysis revealed that participants simultaneously developed their TPACK and SRL strategies through microlearning modules. This confirmed the role of microlearning-based GenAI training in fostering both technological-pedagogical knowledge and self-regulated learning strategies. Early journal entries showed that participants initially focused on basic GenAI functionalities, while later entries demonstrated a sophisticated integration of TPACK knowledge and SRL strategies for lesson planning and instruction.

In the early weeks, participants primarily concentrated on technical competencies and establishing structured learning routines. Fifteen out of 19 participants reported challenges in managing their learning approach. Teacher 5's Week 2 journal entry captured this early stage:

I'm becoming proficient in basic prompt engineering for vocabulary and grammar exercises. The challenge is learning to leverage ChatGPT effectively for more complex tasks like designing communicative activities and reading comprehension materials. So far, the microlearning module has helped me structure my progress. Today, I focused on crafting prompts for reading comprehension tasks, refined my approach through testing, and created materials suitable for my students.

By Week 4, participants began reporting more sophisticated learning strategies. Teacher 12 noted:

I believe established a systematic approach. I start by analyzing the module focus - like today's listening skills development - and then experiment with prompts for various task types and proficiency levels. After testing pre-, during-, and post-listening activities in ChatGPT, I adapt the materials to fit my communicative teaching context. Today, we used Twee for listening activities, allowing me to compare the activities developed with ChatGPT. This step-by-step approach helps me master AI tools and pedagogical applications.

Sixteen participants reported that microlearning helped them develop specific SLR through strategies, demonstrating how structured engagement with GenAI tools enhanced their pedagogical decision-making. Teacher 8 reflected in Week 6:

I maintain a detailed learning log tracking my mastery of AI tools. I know when to use ChatGPT for writing feedback, Magic School for grammar explanations, Perplexity for cultural content, and Twee for differentiated

exercises. By monitoring my progress and combining tool features strategically, I create effective teaching materials.

As participants' became more confident in their SRL approaches, their TPACK development became more evident, reinforcing the connection between structured SRL and pedagogical competence. Teacher 10's Week 8 reflection demonstrated this integration:

My systematic learning approach has helped me effectively use AI tools. I now use ChatGPT for initial differentiated task creation, Perplexity for cultural content, Magic School for visual grammar explanations, and Twee, Magic School, Diffit.me, and ChatGPT for practice activities. I now understand each tool's strengths and combine them to meet my diverse student needs.

4.1.1 Case study: Teacher 1's journey in TPACK and SRL development

Teacher 1's experience illustrates the parallel development of TPACK knowledge and SRL strategies. Her weekly journals and interview responses revealed a transformation from uncertainty to strategic competence in lesson planning and instructional design using GenAI tools.

Week 1 journal focused on initial GenAI foundations:

I'm learning the basics of ChatGPT's interface and simple prompts for vocabulary exercises. While the microlearning modules are manageable, I need a better system for practicing these fundamental skills.

By Week 4, her entries showed emerging competence in both TPACK and SRL strategies, demonstrating the role of microlearning in structuring her learning process: teaching applications:

I have set specific goals for each AI tool and use them systematically to generate level-appropriate materials. Today, I used Alayna.us for lesson planning, Perplexity for researching authentic materials, and Magic School for creating listening comprehension tasks.

Week 8 demonstrated mastery of advanced ELT applications and refined SRL approach:

My systematic approach incorporates multiple AI tools for differentiated instruction. I start with Perplexity research, then use ChatGPT, Twee, and Magic School to create engaging activities.

Weeks 9–12, her reflections indicated sophisticated integration of TPACK knowledge and SRL strategies into her workflow:

I have established a comprehensive workflow for creating assessment materials. I designed rubrics with ChatGPT, refined them with MagicSchool, and checked standards with Perplexity. Next, I used Diffit.me to create level-

specific error correction activities. My reflection log helps me identify the best tool combinations for different teaching objectives.

In her final Week 13 reflection and interview, Teacher 1 demonstrated synthesis of TPACK and SRL strategies for sustained professional development:

The course taught me technical mastery and strategic learning. I created a portfolio of AI-enhanced materials and implementation plans, and developed a systematic approach to evaluating and integrating new AI tools. Each tool has its strengths - ChatGPT for content, Perplexity for research, Magic School for explanations, Twee for practice activities, etc.

These findings highlight how microlearning modules supported both the development of pre-service teachers' TPACK and their ability to self-regulate their learning, confirming the relevance of microlearning for structured teacher learning and instructional design (RQ1).

4.2 Theme 2: Learning and applying GenAI through microlearning experiences

Addressing RQ2, our analysis revealed how pre-service teachers learned and applied GenAI tools through microlearning. They progressed from basic tool familiarization to sophisticated pedagogical integration, with microlearning modules supporting this growth.

Initially, participants focused on understanding fundamental GenAI capabilities, and fourteen reported that the bite-sized format helped them overcome AI anxiety. Teacher 7 noted in Week 2:

Breaking down GenAI learning into small chunks makes it less daunting. Today, I focused on designing communicative activities, learning to prompt ChatGPT, and practicing group tasks with Twee. This targeted approach helped me see how AI can enhance my lesson planning.

By mid-course, participants began reporting more nuanced experiences with GenAI applications. Teacher 13's Week 6 journal demonstrated this evolution:

Microlearning helps me master AI tools for specific teaching tasks. I combine ChatGPT for writing feedback, Wordtune for grammar explanations, and Twee for multilevel practice activities. Each module builds on previous ones, developing my workflow for differentiated language instruction.

Seventeen participants reported that microlearning helped them discover unique pedagogical applications for GenAI. Teacher 4 explained in their interview:

Bite-sized modules helped me discover precise teaching applications for each AI tool. Perplexity finds culturally relevant materials, while Magic School visualizes grammar concepts better. I experimented with unique combinations - like using ChatGPT for lesson frameworks and enhancing them with tool-specific features for differentiation.

The data revealed three key patterns in participants' experiences:

1. Progressive Skill Building

Teacher 15's Week 7 reflection captured this aspect:

Microlearning helped me build skills across different AI tools systematically. I started with ChatGPT for writing feedback, used Perplexity for cultural content research, and learned to combine tools for differentiated instruction. Now, I can create comprehensive, multilevel materials using each tool's strengths.

2. Immediate Application

Teacher 9 noted in Week 5:

Today, I learned to use Magic School for grammar visualization and created differentiated practice activities in Twee. By combining these with ChatGPT's writing feedback features, I developed a complete lesson sequence

3. Iterative Improvement

Teacher 18's interview revealed:

If ChatGPT's writing feedback were not detailed enough, I would try Magic School's approach. If Perplexity's cultural content requires adaptation, I refine it using Twee's level-specific features. This helped me develop expertise in matching tools to specific teaching needs.

4.2.1 Case Study: Teacher 6's learning and application journey

Teacher 6's experience exemplifies the typical progression in learning and applying GenAI through microlearning. Her journal entries tracked this development:

Week 2:

I started with simple prompt writing for vocabulary lists. The five-minute module format helps me focus on mastering one skill at a time.

Week 5:

Now, we are integrating various GenAI features. Today, I utilized a text leveler to create tiered reading texts and then utilized Claude to generate comprehension questions at multiple levels.

Week 8:

Microlearning modules helped me develop a systematic approach to using AI. I can quickly generate, evaluate, and adapt materials for different teaching contexts without feeling overwhelmed.

Final Interview:

In retrospect, the step-by-step approach was crucial. Each module boosted my confidence with GenAI tools. Now, I can seamlessly integrate AI into my lesson-planning workflow. I have established specific procedures for different teaching tasks - from creating authentic dialogues to developing assessment rubrics.

These findings demonstrate how microlearning modules supported pre-service teachers in progressively engaging with and applying GenAI tools in their teaching practice. The structured, iterative learning process allowed them to build confidence, refine pedagogical techniques, and integrate AI-based strategies into real-world instructional planning (RQ2).

4.3 Theme 3: Microlearning's impact on self-regulated GenAI learning

Addressing RQ3, our analysis revealed specific challenges and benefits of using microlearning modules for GenAI integration, particularly with SRL. The data showed how the microlearning format supported and occasionally constrained participants' development of SRL strategies while learning GenAI tools.

Sixteen participants reported that microlearning supported effective goal-setting and time management. Teacher 3's Week 4 journal noted, "Short modules help me set clear learning targets like mastering prompt chaining, and achieve them in one session. This makes my learning more structured and measurable." Self-monitoring and progress tracking were also enhanced by the microlearning format. Teacher 11 reflected in Week 6,

Discrete learning chunks help me monitor my progress. I've created a checklist of AI skills from the modules and can track my growth. When I struggle with a concept, I can quickly review specific modules, revisiting resources like introductory videos and infographic overviews.

Furthermore, Teacher 16's interview revealed that the microlearning format helped them develop a focused learning rhythm, tackling AI tools in small steps and mastering each tool's teaching applications without feeling overwhelmed.

However, thirteen participants initially struggled with the condensed format, finding it challenging to balance depth and brevity. Teacher 19 noted in Week 3,

The five-minute modules introduce key concepts, but I've learned to extend my learning effectively. After a module on ChatGPT's communicative activities, I create additional practice tasks, test them across AI tools, and document the results. These self-directed extensions help me achieve deeper mastery.

Maintaining learning momentum between modules was another challenge. Teacher 14's Week 5 reflection highlighted,

I maintain momentum between modules through structured practice. I dedicate 15 minutes each morning to reinforcing yesterday's skills - like applying ChatGPT's writing feedback features to different student levels or comparing grammar explanations across Magic School and Twee. My learning

journal tracks what works best for specific teaching goals, helping me build systematically on each module."

Participants also found solutions to integrate knowledge from separate modules. Teacher 19 described their approach,

Initially, the separate modules felt fragmented, but creating concept maps helped me integrate different AI applications and teaching strategies. For example, I use ChatGPT to generate initial content, Diffit.me to develop reading and vocabulary activities, and Alayna to create a lesson plan. This helped me see the bigger picture.

4.3.1 Case study: Teacher 2's development of SRL strategies

Teacher 2's journey throughout the course exemplifies how microlearning format facilitated the development of effective SRL strategies for integrating AI tools into teaching practices.

Week 1 Journal:

The modules introduce different AI tools quickly, but I need a system to practice them. I am currently testing digital note-taking to catalogue each tool's features.

Week 4 Journal:

My current routine involves scanning tool objectives, testing features with authentic teaching materials, and documenting AI effectiveness for specific tasks. Today, I compared ChatGPT and Quill for grammar instruction.

Week 7 Journal:

I cluster related AI skills systematically. For assessment design, I combine ChatGPT's rubric creation with MagicSchool's, identifying the best tools for learner assessment, like Quizbot or Mizou, and practicing each thoroughly.

Final Interview:

In the final interview, Teacher 2 reflected on how the microlearning format had taught them a systematic approach to AI integration, explaining,

I now align AI tools with specific teaching needs, test features with actual student work, track which tools excel at different tasks, and combine AI capabilities with proven teaching methods. These strategies help me evaluate and adopt new AI tools effectively.

This reflection demonstrate how microlearning format had a lasting impact on Teacher 2's approach to learning and integrating new AI technologies into their teaching practices.

4.3.2 Evolution of learning strategies

Participants' SRL strategies progressed from basic time management to sophisticated learning approaches throughout the course. Teacher 5's progression exemplifies this:

Week 2:

Following basic modules on how to use ChatGPT's writing feedback features.

Week 5:

Before testing AI feedback tools, I prepare student writing samples and compare ChatGPT's and other tools' approaches.

Week 8:

Systematic approach: identify teaching needs, test relevant AI tools, create sample materials, evaluate effectiveness, and connect with previous tool combinations.

4.3.3 Impact on future learning

Teacher 17 reflected on the lasting effects on participants' approach to learning new technologies:

This format taught me to master any AI tool methodically. I analyze its teaching applications, test key features with real materials, and gradually integrate successful approaches into my lesson planning.

This theme shows how microlearning helped pre-service teachers develop robust SRL strategies for GenAI integration, including goal setting, reflection, and adaptive learning approaches. Despite initial challenges, participants leveraged microlearning to structure their learning processes and refine their AI-assisted teaching strategies, reinforcing its effectiveness in supporting SRL development (RQ3).

5 Discussion

This study highlights how microlearning can enhance pre-service teachers' TPCK and SRL strategies in GenAI-focused learning environments. The findings underscore the importance of a phased approach in teacher training programmes, starting with basic TK and TCK and then progressing to the comprehensive application of TPCK in designing learning tasks and materials. This layered progression aligns with prior research on the incremental development of teacher competencies in technology-enhanced learning environments (Koehler & Mishra, 2009; Kohnke et al., 2024a, b).

Our findings align with Kohnke et al. (2024a), who demonstrated that microlearning fosters the development of teachers' professional digital competence. Similarly,

Shamir-Inbal and Blau (2022) highlighted the pivotal role of self-regulation in ICT teacher leaders' professional growth through microlearning. Consistent with Hosseini et al. (2020), our study further reveals that GenAI-focused microlearning supports teachers in cultivating self-regulation skills. Together, these studies suggest a nuanced relationship between microlearning and self-regulation, warranting further investigation. While prior research has predominantly emphasized microlearning's efficiency, our study underscores its dual capacity to enhance both TPACK and SRL through AI-driven contextualization. Extending Fidan's (2023) exploration of microlearning-supported flipped classrooms and their effects on pre-service teachers' performance, motivation, and engagement, our findings demonstrate that microlearning also contributes to the development of teachers' self-regulation. This adds empirical support to Alias and Razak's (2025) hypothesis that microlearning positively influences self-regulation, learning performance, motivation, and engagement.

The study reveals a significant insight: the dynamic and iterative nature of the learning process. Different microlearning tasks emphasise various aspects of TPACK (e.g. TCK, TPK) or their integrated application, while also requiring diverse levels of pedagogical and content knowledge. This flexibility allowed participants to tailor their learning paths based on their preferences and needs, illustrating the value of adopting a 'learning by doing' approach in teacher training. This adaptability not only enhanced the learning experience but also fostered personalised skill development, demonstrating microlearning's effectiveness in supporting different learner profiles.

The participants' iterative refinement of SRL strategies further emphasises the scaffolded nature of microlearning. Initially, participants focused on time management and goal-setting, gradually progressing to advanced self-monitoring and strategic adaptation. The incremental skill-building facilitated by microlearning proved instrumental in reducing AI-related anxiety. Additionally, it supported the practical application of tools, aligning with previous research on the role of structured learning experiences in technology adoption (Chiu, 2024; Moorhouse et al., 2024). This interplay between SRL and TPACK demonstrates the need for teacher training programmes to align these domains, ensuring educators can effectively integrate AI tools into instructional design.

The study also revealed a transition among participants: from focusing on isolated tool functionalities to integrating multiple tools into cohesive teaching workflows. This progression reflects microlearning's capacity to bridge the gap between theoretical knowledge and practical application. By experimenting with, adapting and refining their use of GenAI tools, pre-service teachers gained confidence and proficiency in incorporating these technologies into various teaching contexts. To harness this potential, teacher education programmes should incorporate task-oriented microlearning modules that emphasise immediate application, iterative refinement and the development of cohesive pedagogical strategies.

Despite its advantages, the study identified several challenges associated with microlearning. Participants reported issues with the brevity of modules, difficulty maintaining learning momentum and challenges in synthesising fragmented knowledge. These findings suggest that complementing microlearning with reflective practices – such as maintaining learning logs and creating concept

maps – could enhance knowledge integration and promote sustained engagement. Moreover, incorporating opportunities for collaboration and feedback through peer discussions, group projects and mentoring sessions can support pre-service teachers in synthesising knowledge and sustaining motivation. Providing curated resources and ongoing support can further mitigate microlearning's brevity-related challenges.

Nevertheless, while GenAI tools enhance content creation, they risk fostering over-reliance, a concern corroborated by Schoeffler et al. (2025) and Zhai et al. (2024), who warn that excessive dependency on AI may erode foundational curriculum design competencies. Our findings echo this tension: a subset of participants struggled to conceptualize lesson plans without AI assistance, underscoring the need for balanced integration strategies. To preserve pedagogical autonomy, we propose pairing AI tools with 'unplugged' pedagogical tasks such as analog lesson planning workshops or peer-led content critiques to ensure teachers retain and refine core instructional skills.

Compounding these risks is the 'black-box' nature of AI-driven systems, which, as highlighted by Hassija et al. (2024) and Adadi and Berrada (2018), generate powerful predictions but lack transparent rationale. This opacity may inadvertently stifle teachers' ability to critically evaluate or adapt AI-generated content, a limitation observed in our study where participants often accepted algorithmic suggestions uncritically. For instance, several teachers could not articulate the rationale behind AI-recommended activity sequences. To address this, we advocate for the importance of fostering critical awareness of GenAI tools among educators. Teacher education programmes should incorporate AI-literacy components to equip teachers with the skills to critically evaluate AI-generated materials, address biases and promote equitable and inclusive teaching practices. By integrating these elements, teacher training programmes can prepare educators to harness the full potential of GenAI tools while addressing their ethical and pedagogical implications.

6 Conclusion

This study highlights the effectiveness of microlearning modules in developing pre-service teachers' TPCK and SRL, enabling them to integrate GenAI tools into their teaching practices. By addressing the dual development of these competencies and offering practical insights into overcoming the associated challenges, this study contributes to the growing body of research on integrating AI into education. It offers a scalable, structured approach that teacher education programmes can adopt to equip future educators with the skills to navigate AI-enhanced classrooms. In doing so, it makes a significant contribution to teacher training. It bridges the gap between AI literacy, SRL, and pedagogical applications, ensuring that pre-service teachers are prepared for the evolving demands of future classrooms.

This study also proposes a structured framework for integrating microlearning into teacher education that combines phased skill development, scaffolded engagement with AI tools, and iterative, self-regulated learning practices. The framework has three key components: (1) a progressive phase-based structure that guides users

from foundational to advanced applications of AI tools in teaching; (2) an iterative learning cycle through which pre-service teachers engage in microlearning modules, apply AI tools in lesson planning, and reflect on their strategies; and (3) an SRL support system that incorporates goal-setting, practice-based refinement, and peer collaboration. By following this framework, teacher education programmes can provide a replicable model that systematically cultivates AI literacy and pedagogical competencies, ensuring that pre-service teachers develop the technical proficiency and reflective teaching practices necessary for success in AI-enhanced learning environments.

This research has several limitations that suggest valuable directions for future studies. First, the study comprised a limited dataset of 19 pre-service teachers, although it still allowed for a “new and richly textured understanding” (Sandelowski, 1995, p. 183) of the phenomenon under study. In line with the nature of qualitative research, readers are encouraged to position themselves vis-à-vis the results and determine the relevance of the findings to their contexts (Merriam & Tisdell, 2016). Future research could employ larger samples and mixed-methods approaches, incorporating performance-based assessments or classroom observations to enhance generalisability and capture a broader range of experiences. Additionally, longitudinal approaches could examine the sustainability of AI use in teaching and whether pre-service teachers maintain or enhance their competencies over time.

Participants self-selected into this elective course, so there may have been selection bias; those more open to technology integration may have been more likely to enrol. However, the participants’ varied digital literacy backgrounds and minimal prior exposure to GenAI tools reduced this risk, as did the fact that the study explored their learning process rather than their pre-existing expertise. All of the participants were based in Hong Kong, so the results reflect the perspectives of pre-service teachers within this specific cultural and educational context. Future research including participants from different countries and regions could facilitate comparative analyses, providing insights into cross-cultural similarities and differences in TPCK and SRL development.

Future studies could also explore tailoring microlearning to in-service teachers, examining its long-term impact on their teaching efficacy and student learning outcomes. A comparative analysis between pre-service and in-service teachers would provide valuable insights into how different teaching environments, challenges, and professional experiences influence the development of TPCK and SRL strategies. This research could identify the unique approaches and needs of experienced educators, optimising microlearning’s effectiveness across varied teaching contexts.

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Declarations

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